

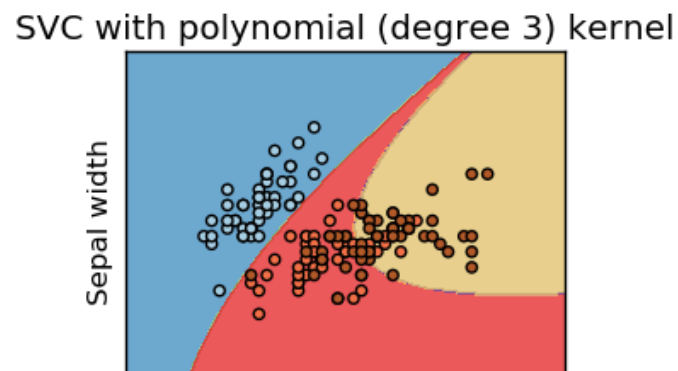
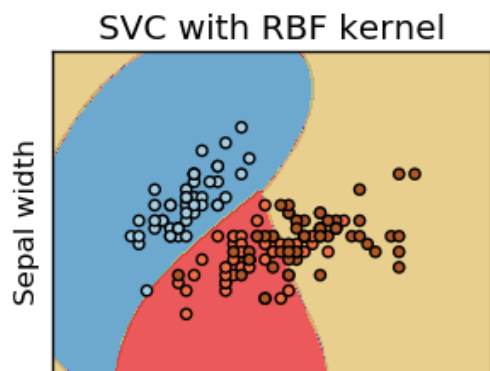
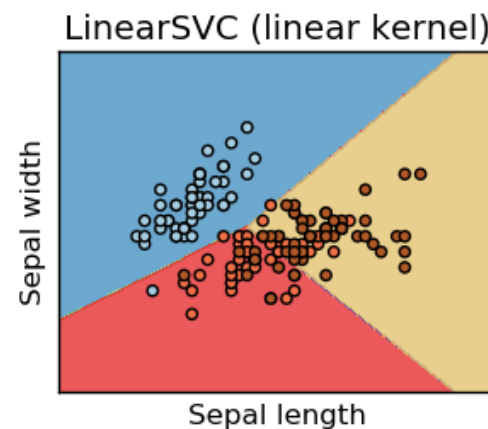
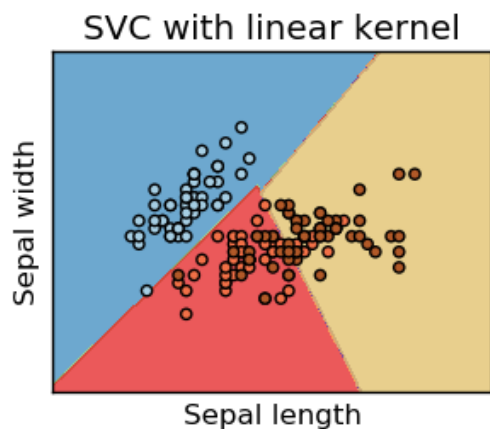
In Depth: Parameter tuning for SVC

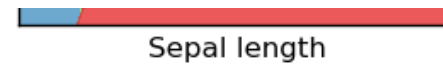
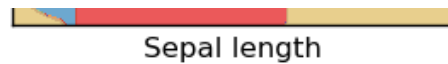


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In this post we will explore the most important parameters of Sklearn SVC classifier and how they impact our model in term of overfitting.

Support Vectors Classifier tries to find the best hyperplane to separate the different classes by maximizing the distance between sample points and the hyperplane.

We will use the Iris data from sklearn.

We start by loading the libraries and SVC

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn import svm, datasets
%matplotlib inline

# import some data to play with
iris = datasets.load_iris()
X = iris.data[:, :2] # we only take the first two features. We could
# avoid this ugly slicing by using a two-dim dataset
y = iris.target
```

We use the same plotting script for our plots.

```
def plotSVC(title):
    # create a mesh to plot in
    x_min, x_max = X[:, 0].min() - 1, X[:, 0].max() + 1
    y_min, y_max = X[:, 1].min() - 1, X[:, 1].max() + 1
    h = (x_max / x_min)/100
    xx, yy = np.meshgrid(np.arange(x_min, x_max, h),
                          np.arange(y_min, y_max, h))

    plt.subplot(1, 1, 1)
    Z = svc.predict(np.c_[xx.ravel(), yy.ravel()])
    Z = Z.reshape(xx.shape)
    plt.contourf(xx, yy, Z, cmap=plt.cm.Paired, alpha=0.8)

    plt.scatter(X[:, 0], X[:, 1], c=y, cmap=plt.cm.Paired)
    plt.xlabel('Sepal length')
    plt.ylabel('Sepal width')
    plt.xlim(xx.min(), xx.max())
    plt.title(title)
    plt.show()
```

Kernel

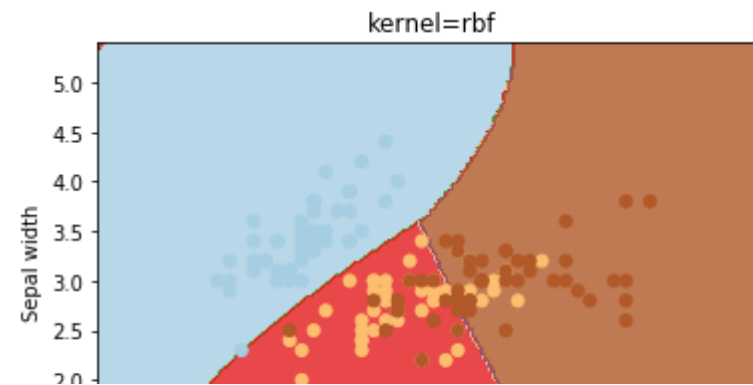
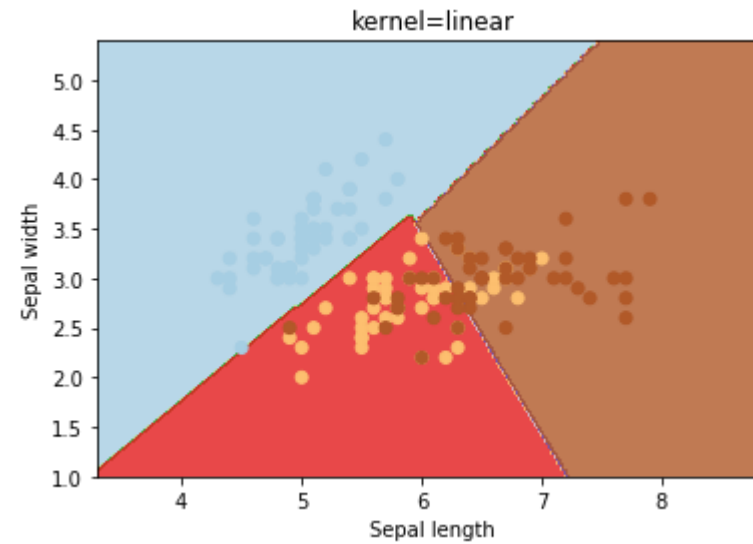
kernel parameters selects the type of hyperplane used to separate the data.

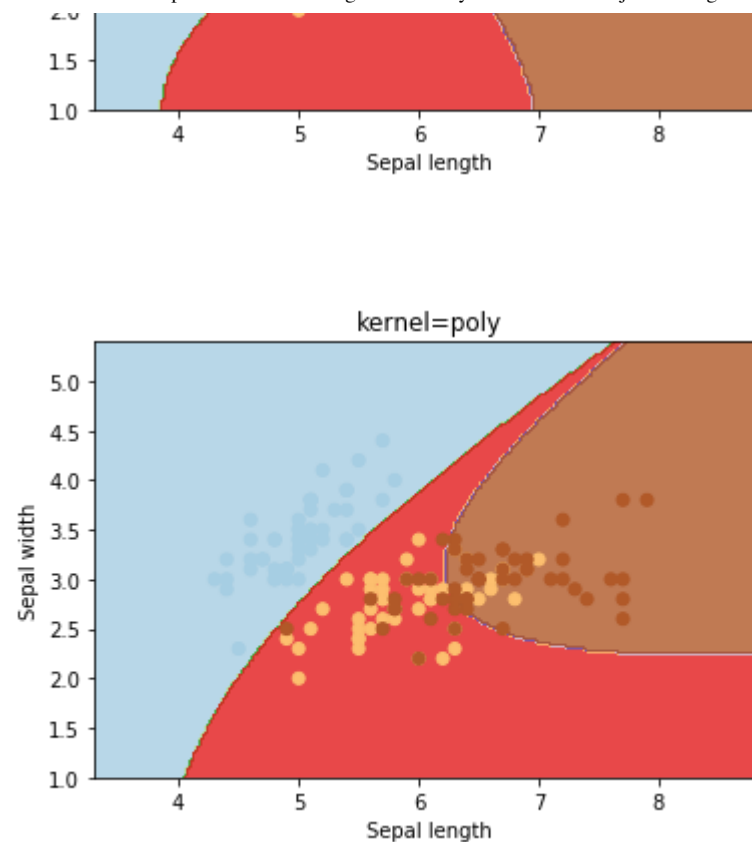
Using 'linear' will use a linear hyperplane (a line in the case of 2D data).

'rbf' and 'poly' uses a non linear hyper-plane

```
kernels = ['linear', 'rbf', 'poly']
```

```
for kernel in kernels:  
    svc = svm.SVC(kernel=kernel).fit(X, y)  
    plotSVC('kernel=' + str(kernel))
```



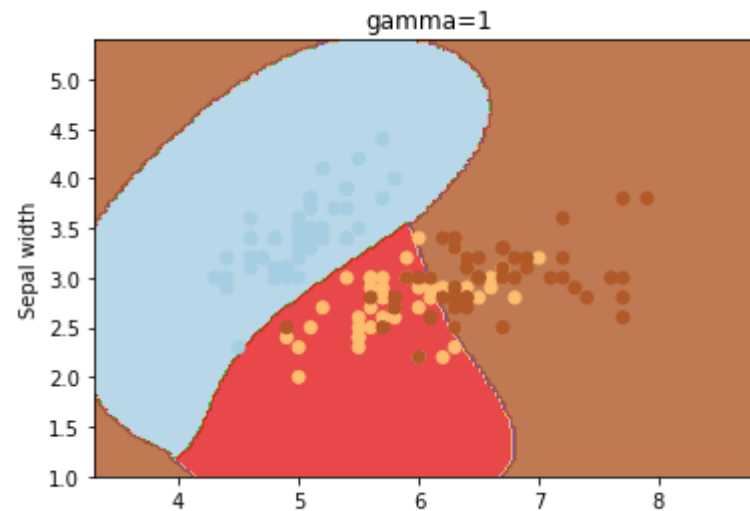
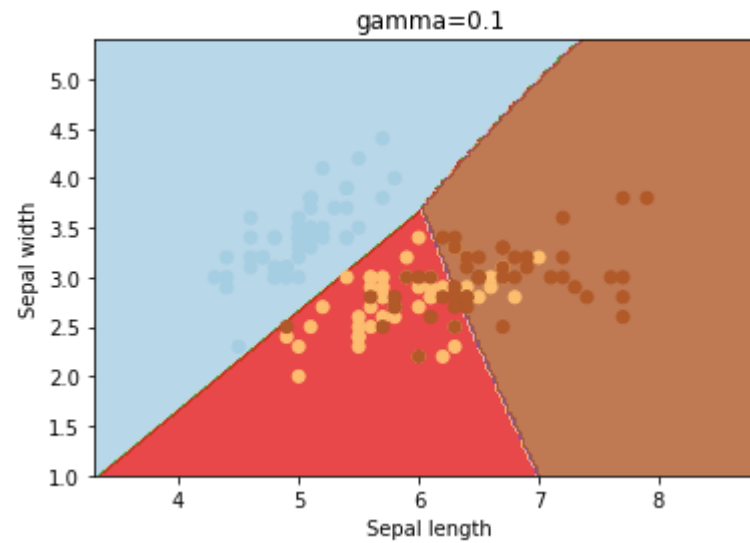


gamma

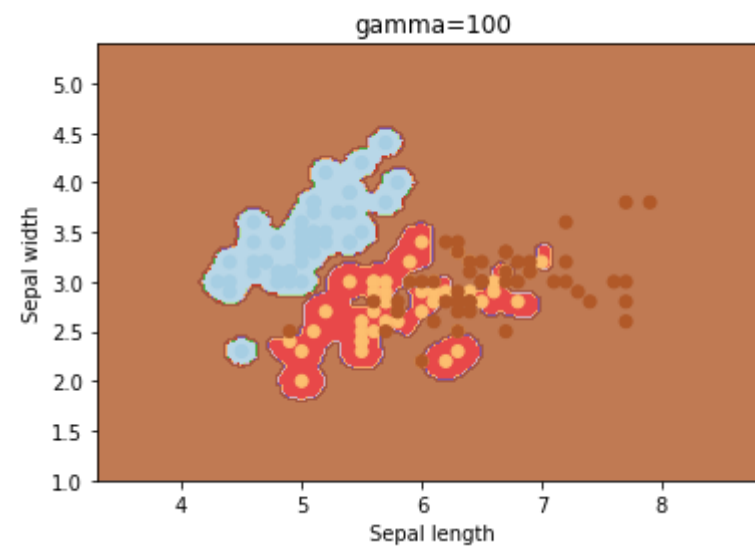
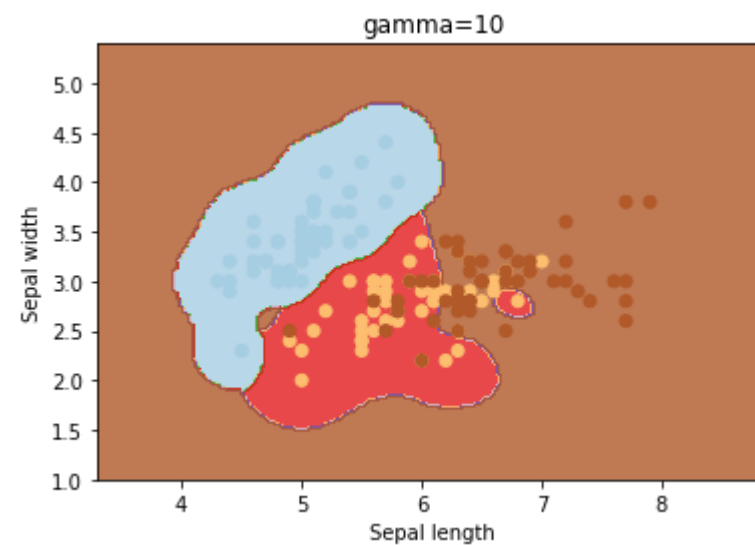
gamma is a parameter for non linear hyperplanes. The higher the gamma value it tries to exactly fit the training data set

```
gammas = [0.1, 1, 10, 100]
```

```
for gamma in gammas:  
    svc = svm.SVC(kernel='rbf', gamma=gamma).fit(X, y)  
    plotSVC('gamma=' + str(gamma))
```



Sepal length



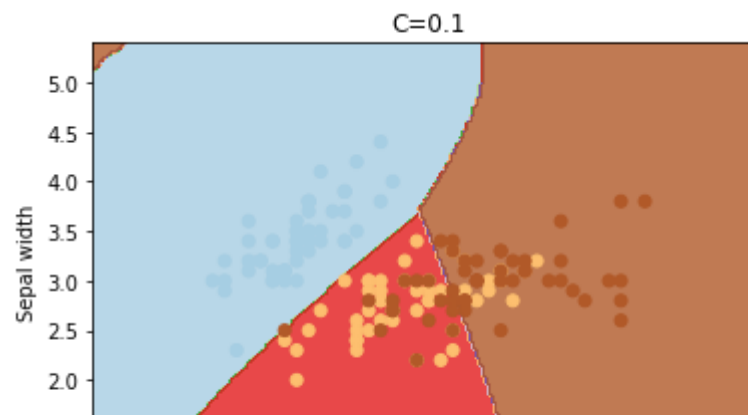
We can see that increasing gamma leads to overfitting as the classifier tries to perfectly fit the training data

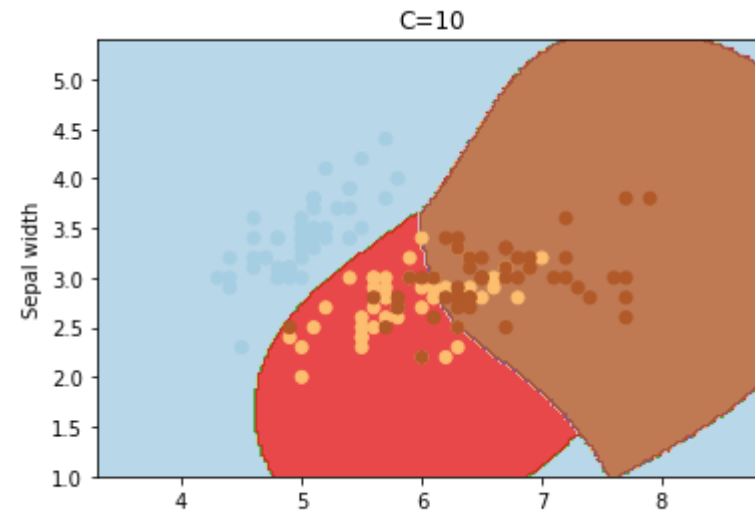
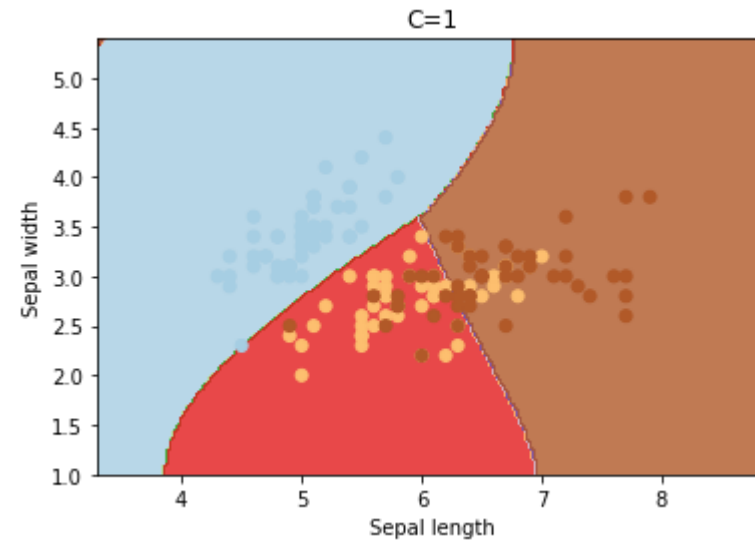
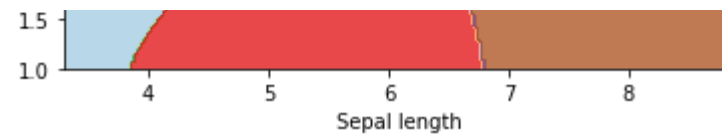
C

C is the penalty parameter of the error term. It controls the trade off between smooth decision boundary and classifying the training points correctly.

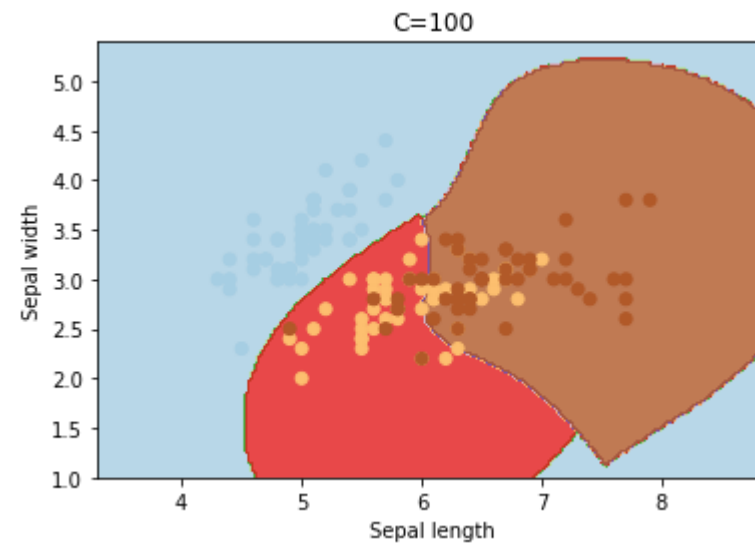
```
cs = [0.1, 1, 10, 100, 1000]
```

```
for c in cs:  
    svc = svm.SVC(kernel='rbf', C=c).fit(X, y)  
    plotSVC('C=' + str(c))
```





Sepal length



Increasing C values may lead to overfitting the training data.

degree

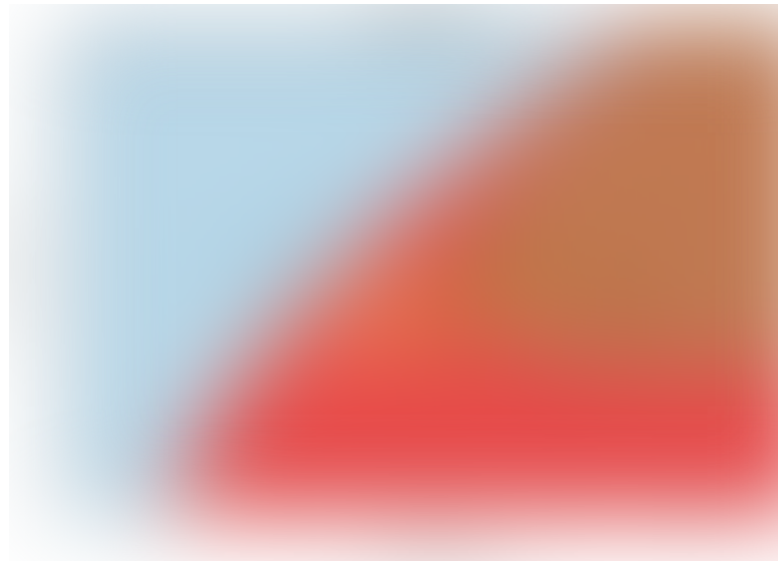
degree is a parameter used when kernel is set to 'poly'. It's basically the degree of the polynomial used to find the hyperplane to split the data.

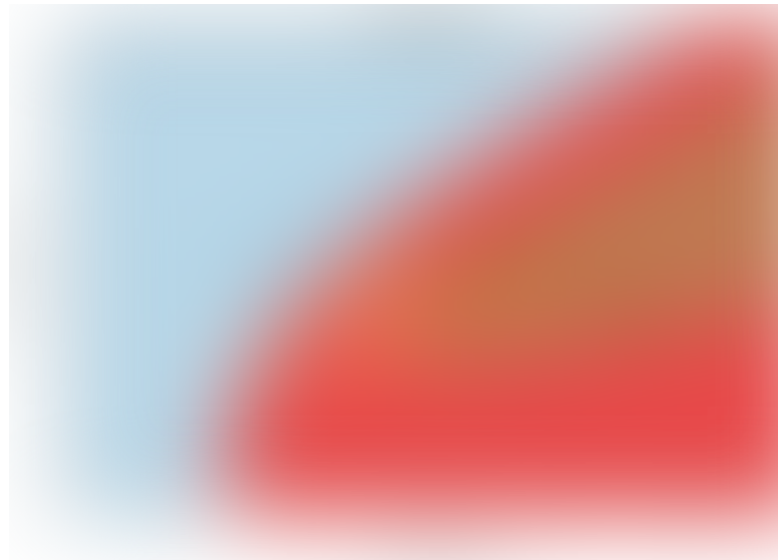
```
degrees = [0, 1, 2, 3, 4, 5, 6]
```

```
for degree in degrees:  
    svc = svm.SVC(kernel='poly', degree=degree).fit(X, y)  
    plotSVC('degree=' + str(degree))
```









Using $\text{degree}=1$ is the same as using a 'linear' kernel. Also, increasing this parameters leads to higher training times.

• • •

The inDepth series investigates how model parameters affect performance in term of overfitting and underfitting. You can check parameter tuning for tree based models like Decision Tree , Random Forest, Gradient Boosting and KNN.

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